ELEC60029 Semiconductor Devices- 2024: *Answers:*

Q1

(a)
Generation rate: $G(T) = f_1(T)$
Recombination rate: $R(T)=f_2(T) \times n \times p$
In equilibrium: $R(T)=G(T)$
Hence, $f_2(T) \times n \times p = f_1(T)$
This gives:

$$n \times p = \frac{f_1(T)}{f_2(T)} = constant$$

$$\therefore n \times p = n_i^2$$

[4]

(b)
Conductivity $\sigma = e(n\mu_n + p\mu_p)$
Here, n and p are determined by the doping. Sample 1 is intrinsic, hence $n = p = n_i$, Sample 2 has compensated doping with $n = \underline{N_D} >> N_A$. Sample 3 and 4 have $n = \underline{N_D}$ and $p = \underline{N_A}$ respectively. Finally, in Si, $\mu_n > \mu_p$. This gives:

$$\sigma_2 > \sigma_3 > \sigma_4 > \sigma_1$$

[4]

(c)
(i) When the base current is sufficiently large, it will build up a minority carrier charge in the base that is larger than the charge that causes saturation of the collector current. At saturation, the collector current is no longer controlled by the current gain β.

We then have $I_c = \beta I_B$ as this is limited by the load, $I_{c,sat} = V_{CC} / R$ where V_{CC} is the EC bias and R the load resistance.

[4]

ii) t_{rise} is the time it takes the base charge to reach the point where the collector current saturates:
$Q_B = \tau_t I_{c,sat} = \tau_t V_{CC} / R$. Here Q_B keeps increasing but I_C saturates. τ_t is the base saturation time.

[4]

(d) The Fermi-Dirac *f(E)* distribution (see formula sheet) is given by:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

In contrast, the Boltzmann *f'(E)* distribution (this is taught in lectures) is given by:

$$f'(E) = \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right)}$$

The difference between the two distributions for <5% error is then:

ANSWERS

$$\frac{f'(E) - f(E)}{f(E)} < 0.05$$

[2]

This implies that:

$$\frac{f'(E)}{f(E)} - 1 < 0.05$$
$$\Rightarrow \frac{f'(E)}{f(E)} < 1.05$$

$$\Rightarrow \frac{1 + exp\left(\frac{E - E_F}{k_B T}\right)}{exp\left(\frac{E - E_F}{k_B T}\right)} < 1.05$$

$$\Rightarrow \frac{1}{exp\left(\frac{E - E_F}{k_B T}\right)} < 0.05$$

$$\Rightarrow exp\left(\frac{E - E_F}{k_B T}\right) > 20$$

$$\Rightarrow E - E_F > k_B T ln20 = 0.026 \times 2.99 = 0.078\ eV$$

[4]

(e)

i) Forward bias, near turn-on voltage: Diffusion

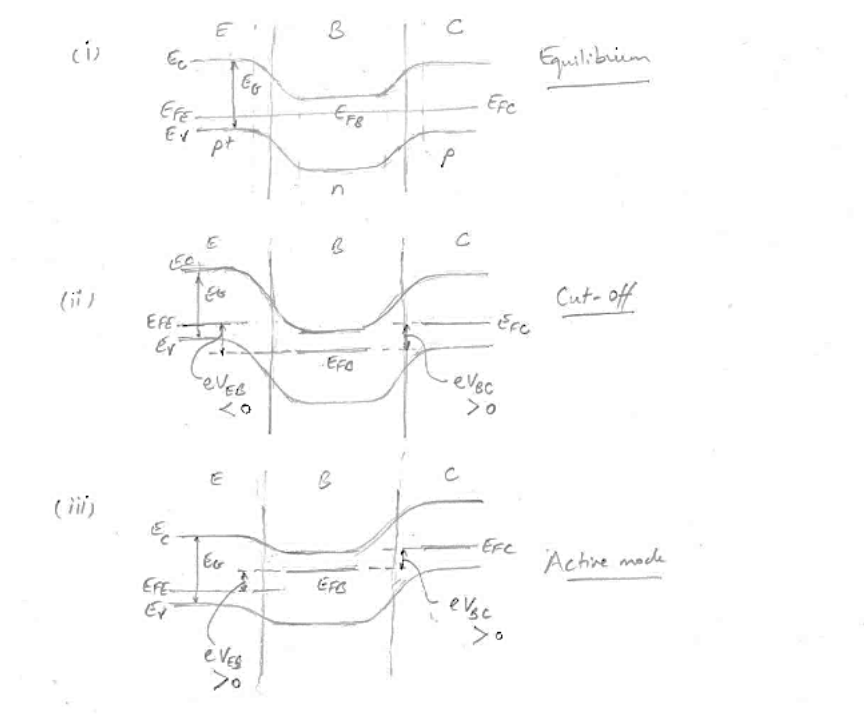
ii) Forward bias, near zero bias: Diffusion, recombination in depletion region (low injection condition)

iii) Reverse bias, well away from breakdown, within depletion region. Minority carrier drift. There may also be some generation but reasonable to neglect this at lower reverse bias.

[2+2+2]

ANSWERS

(f)

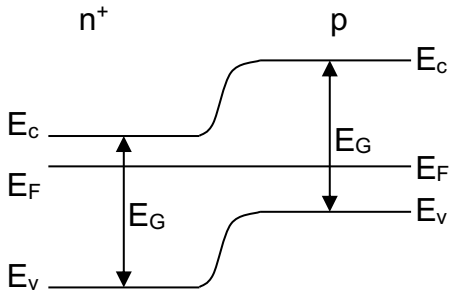

(i)
E
B
C
Equilibrium
E_C
E_G
E_{FE}
E_V
E_{FB}
E_{FC}
p+
n
p
(ii)
Cut-off
eV_{EB} < 0
eV_{BC} > 0
(iii)
Active mode
eV_{EB} > 0
eV_{BC} > 0

[4 + 4 + 4]

ANSWERS

Q2

2. a) i)

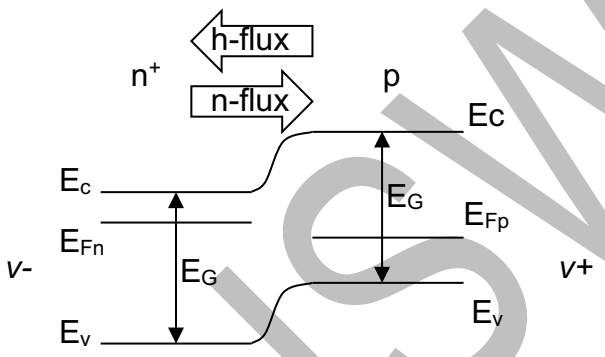


[6]

Here,
1. E_F needs to be constant.
2. for *n*-type $E_c - E_F < E_F - E_v$; for *p*-type $E_c - E_F > E_F - E_v$
3. $(E_c - E_F)_n < (E_F - E_v)_p$ (the distance between Fermi level and conduction band in the *n*-layer needs to be smaller than the distance between Fermi level and valence band in the *p*-type layer because *n* is more heavily doped than *p*.
4. the bandgap in *n* and *p* type layers are the same.

It is not necessary to draw the relative depletion widths to scale.

ii)



[6]

Important rules:
5. E_F not constant. E_{Fn} should be higher than E_{Fp}.
6. Rule 2 should still be applied (no change with previous diagram)
7. Rule 3: $(E_c - E_F)_n < (E_F - E_v)_p$ should still be applied (no change with previous diagram)
8. The bandgap in *n* and *p* type layers are the same, as previously.
9. The potential barrier between the *n* and *p* layers should be smaller than in the previous diagram.

It is not necessary to draw the relative depletion widths to scale.

b) From the formulae list copy.

The electron current density of the *pn* diode: $J_n = \frac{eD_n n_{p_0}}{L_n}\left(e^{\frac{eV}{kT}} - 1\right)$ and hole current density of the *pn* diode: $J_p = \frac{eD_p p_{n_0}}{L_p}\left(e^{\frac{eV}{kT}} - 1\right)$

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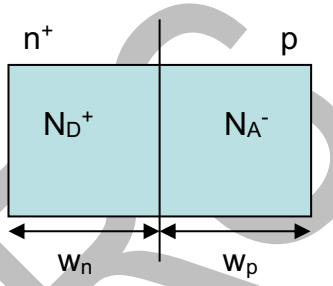
Rewrite the expression as a function of doping density:

$$J_n = \frac{eD_n n_i^2}{L_n N_A}\left(e^{\frac{eV}{kT}} - 1\right) \;\&\; J_p = \frac{eD_p n_i^2}{L_p N_D}\left(e^{\frac{eV}{kT}} - 1\right)$$

Since $\frac{D_n}{L_n} > \frac{D_p}{L_p}$ and $N_A << N_D$, $J_n > J_p$. Hence, electron current density is larger than hole current density.

[4]

c) If doping density and voltage cannot be changed then the only parameter remaining is the length of the *n* or *p* region.



$$J_{new} = I_{0_{new}}\left(e^{\frac{eV}{kT}} - 1\right) = 10 \times J_{old} = 10 \times \left(\frac{eD_n n_i^2}{L_n N_A} + \frac{eD_p n_i^2}{L_p N_D}\right)\left(e^{\frac{eV}{kT}} - 1\right)$$

$$I_{0_{new}} = 10 \times \left(\frac{eD_n n_i^2}{L_n N_A} + \frac{eD_p n_i^2}{L_p N_D}\right)$$

$$\left(\frac{eD_n n_i^2}{X_p N_A} + \frac{eD_p n_i^2}{X_n N_D}\right) = 10 \times \left(\frac{eD_n n_i^2}{L_n N_A} + \frac{eD_p n_i^2}{L_p N_D}\right)$$

Definition of parameters: X_p is the length of the *p*-type layer, X_n is the length of the *n*-type layer. Remember holes diffuse in the n-type layer and electrons diffuse in the *p*-type layer.

As $N_D = 20000 N_A$, $D_n = 2\ D_p$ and $L_p = 10 L_n$, the terms with N_D in the denominator may be neglected. This gives:

$$\frac{eD_n n_i^2}{X_p N_A} = 10 \times \left(\frac{eD_n n_i^2}{L_n N_A}\right)$$

If we now keep the length of the *n*-layer unchanged and change only the length of the *p*-layer X_p then:

$$\left(\frac{eD_n n_i^2}{X_p N_A}\right) = 10 \times \left(\frac{eD_n n_i^2}{L_n N_A}\right)$$

$$X_p = \frac{L_n}{10} = 10^{-4} cm = 1 \mu m$$

Hence, a ten times increase in current can be obtained by a ten times decrease in the length of the *p*-layer or more accurately by making the length of the *p*-layer 10 time smaller than the electron diffusion length. This is a realistic solution.

[4]

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If we would have chosen to keep the length of the *p*-layer unchanged thus X_p remains L_n (X_p=L_n) and change only the length of the *n*-layer X_n then:

$$\left(\frac{eD_n n_i^2}{L_n N_A}+\frac{eD_p n_i^2}{X_n N_D}\right)=10\times\left(\frac{eD_n n_i^2}{L_n N_A}+\frac{eD_p n_i^2}{L_p N_D}\right)$$

$$\left(\frac{eD_p n_i^2}{X_n N_D}\right)=10\times\left(\frac{eD_n n_i^2}{L_n N_A}+\frac{eD_p n_i^2}{L_p N_D}\right)\approx 10\times\left(\frac{eD_n n_i^2}{L_n N_A}\right)\quad [N_D >> N_A]$$

$$X_n=\frac{D_p N_A L_n}{10 D_n N_D}=2.5\times 10^{-6}L_n=2.5\times 10^{-9}cm$$

This would mean that the width of the *n*-type layer should have to become smaller than an atomic distance, thus this is not a realistic solution.

The above is an approximate solution as short diode equations are used. These start to break down when the lengths are equal to the diffusion lengths. Thus, while these equations are extremely good approximations for short layers, they overestimate the current density in longer layers.

[2]

d)

At $t = 0^-$, diode current, in forward bias, is $I_F = V/R$ (assuming diode voltage $v_d << V$).

At $t = 0^+$, diode current, in reverse bias, is $-I_R = -V/R$, determined by the external circuit. Here, I_R is the current magnitude, and the negative sign emphasises the reverse current.

[2]

Using the given equation:

$$i_p(t)=\frac{Q_p(t)}{\tau_p}+\frac{dQ_p(t)}{dt}$$

$$@t=0^-;i(0^-)=I_F\rightarrow Q(0^-)=\tau_p I_F$$
$$@t=0^+;i(0^+)=-I_R\rightarrow Q(0^+)\neq -\tau_p I_R!$$
$$@t=t_{sd};Q(t_{sd})=0$$

$$0<t<t_{sd}\rightarrow i_p(t)=-I_R$$
$$-I_R=\frac{Q_p(t)}{\tau_p}+\frac{dQ_p(t)}{dt}$$

[2]

Solving this equation:

$$-I_R = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$$

$$\frac{-\tau_p I_R - Q_p(t)}{\tau_p} = \frac{dQ_p(t)}{dt}$$

$$\frac{-dt}{\tau_p} = \frac{dQ_p(t)}{\left(\tau_p I_R + Q_p(t)\right)}$$

integrate

$$\frac{-t}{\tau_p} = \ln\left(\tau_p I_R + Q_p(t)\right)\Big|_0^t$$

$$\frac{-t}{\tau_p} = \ln\left(\tau_p I_R + Q_p(t)\right) - \ln\left(\tau_p I_R + Q_p(0)\right)$$

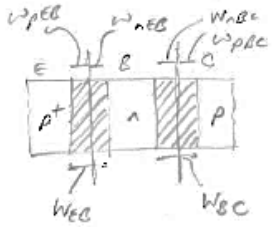
$$\exp\left(\frac{-t}{\tau_p}\right) = \frac{\left(\tau_p I_R + Q_p(t)\right)}{\left(\tau_p I_R + \tau_p I_F\right)}$$

$$Q_p(t) = -\tau_p I_R + \left(\tau_p I_R + \tau_p I_F\right)\exp\left(\frac{-t}{\tau_p}\right)$$

[4]

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Q3:



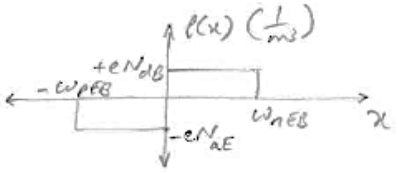
a) Within the EB depletion region, width W_{EB}, charge neutrality implies:

$$eN_{aE}W_{pEB} = eN_{dE}W_{nEB}$$

Here, we assume that dopant atoms are completely ionised and hole/electron carrier concentrations, p, n, are neutralised. As the Fermi level is well away from E_c, E_V in the depletion region, the later assumption is justified.

[4]

b)

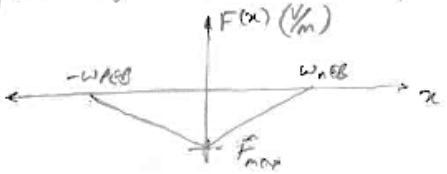


[4]

Nb: Here, the heights of ρ(x) on left/right are not shown to scale for simplicity ($N_a > N_d$)

c) 1-D Poisson Eq:- $\frac{dE}{dx} = \frac{\rho(x)}{\epsilon} = \frac{\rho(x)}{\epsilon_s \epsilon_0} \Rightarrow F = \int \frac{\rho(x)}{\epsilon_s \epsilon_0} dx$

Integration of the distribution of ρ(x) given in (b) $\Rightarrow$



Nb: Here, the width of the distribution on the left/right is shown to scale

[3]

Furthermore, F_{max} is given by the area with the 'rectangle' formed by ρ(x), between $-W_{pEB} \rightarrow 0$.

$$\Rightarrow F_{max} = \frac{-eW_{pEB}N_{aE}}{\epsilon_s \epsilon_0} = \frac{-eW_{nEB}N_{dB}}{\epsilon_s \epsilon_0}$$

[3]

d) To relate electric field to the built-in potential V_0 across the EB junction, we use:

$$\frac{dV}{dx} = -F(x) \Rightarrow V = -\int F(x)\,dx \qquad [2]$$

The integration corresponds to the area under the distribution of $F(x)$ in (c). Considering the triangle formed between $x = -w_{pEB} \rightarrow x = +w_{nEB}$,

$$V_0 = \frac{1}{2} \cdot W_{EB} \cdot F_{max} = \frac{1}{2}(w_{pEB} + w_{nEB}) \cdot F_{max}$$

Furthermore, $W_{EB} = w_{pEB} + w_{nEB} = \frac{\varepsilon_{Si}\varepsilon_0 F_{max}}{e N_{aE}} + \frac{\varepsilon_{Si}\varepsilon_0 F_{max}}{e N_{dB}}$

$$\Rightarrow F_{max} = \frac{e}{\varepsilon_{Si}\varepsilon_0}\left(\frac{N_{aE} N_{dB}}{N_{aE} + N_{dB}}\right) W_{EB} \qquad [4]$$

Substituting for F_{max} in the equation for V_0 above:

$$V_0 = \frac{1}{2} W_{EB} F_{max} = \frac{e}{2\varepsilon_{Si}\varepsilon_0}\left(\frac{N_{aE} N_{dB}}{N_{aE} + N_{dB}}\right) W_{EB}^2$$

$$\Rightarrow W_{EB} = \sqrt{\frac{2\varepsilon_{Si}\varepsilon_0}{e}\left(\frac{N_{aE} + N_{dB}}{N_{aE} N_{dB}}\right) V_0}$$

Finally $+V_{EB}$ forward biases the E-B junction, hence reducing the influence of V_0. This gives the required expression:

$$W_{EB} = \sqrt{\frac{2\varepsilon_{Si}\varepsilon_0}{e}\left(\frac{N_{aE} + N_{dB}}{N_{aE} N_{dB}}\right)(V_0 - V_{EB})} \qquad [4]$$

e) The emitter current is determined by the undepleted width of the base, $W_B' = W_B - (w_{nEB} + w_{nBC})$, and the diffusion current equation:

$$I_E = \frac{e A p_{n0} D_p}{W_B'}\left(\exp\left(\frac{e V_{EB}}{kT}\right) - 1\right)$$

To find W_B', we need to find w_{nEB} and w_{nBC}.

From the rubric:

$$W_{nEB} = \sqrt{\frac{2\epsilon_0\epsilon_{Si}}{e}\cdot\frac{N_{aE}}{N_{aE}N_{dB}+N_{dB}^2}\,(V_0 - V_{EB})}, \quad V_0 = \frac{kT}{e}\ln\left(\frac{N_{aE}N_{dB}}{n_i^2}\right)$$

$$= \sqrt{\frac{2\times 11\times 8.85\times 10^{-14}}{1.6\times 10^{-19}}\cdot\frac{10^{17}}{10^{17}\cdot 10^{16}+(10^{16})^2}\,(0.76-0.5)}$$

$$\Rightarrow V_0 = 0.026\ln\left(\frac{10^{17}\cdot 10^{16}}{(1.45\times 10^{10})^2}\right) = 0.76\ V$$

$$= \sqrt{(12.17\times 10^{6})(9.09\times 10^{-17})(0.26)}$$

$$= 16.96\times 10^{-6}\ cm \approx 0.170\ \mu m.$$

Similarly, $W_{nBC} = \sqrt{\frac{2\epsilon_0\epsilon_{Si}}{e}\cdot\frac{N_{aC}}{N_{aC}N_{dB}+N_{dB}^2}\,(V_0 - V_{CB})}$

$$V_0 = \frac{kT}{e}\ln\left(\frac{N_{aC}N_{dB}}{n_i^2}\right) = 0.026\ln\left(\frac{10^{16}\cdot 10^{16}}{(1.45\times 10^{10})^2}\right) = 0.7\ V$$

$$= \sqrt{(12.17\times 10^{6})\cdot\frac{10^{16}}{10^{16}\cdot 10^{16}+(10^{16})^2}\cdot(0.7+0.5)}$$

Active mode $\Rightarrow V_{BC}$ is reverse biased!

$$= 27\times 10^{-6}\ cm = 0.27\ \mu m. \qquad (3)$$

$$\therefore W_B' = 2 - 0.27 - 0.17 = 1.56\ \mu m. = 156\times 10^{-6}\ cm$$

$$\therefore I_E = \frac{eAp_{n0}}{W_B'}\left(\exp\left(\frac{eV_{EB}}{kT}\right)-1\right)$$

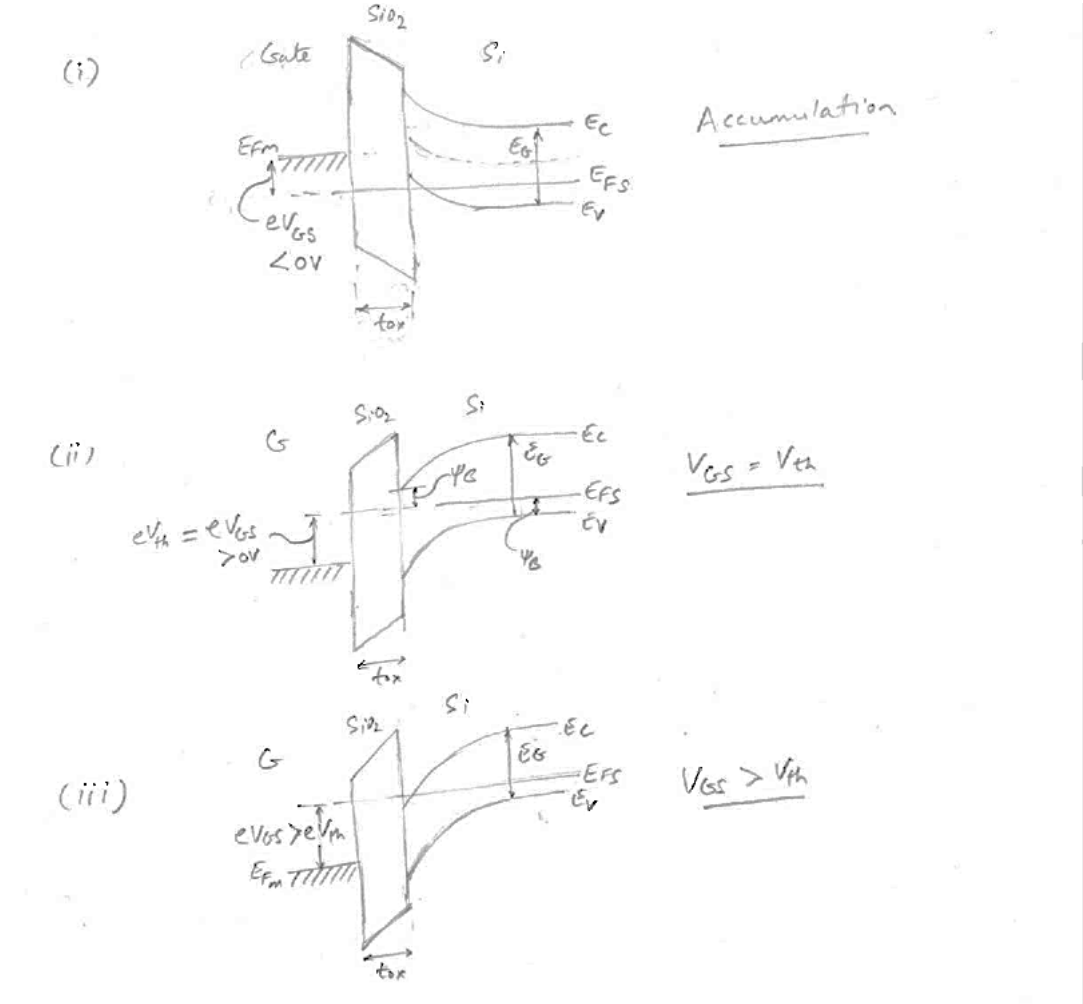
$$= \frac{eA\cdot n_i^2 D_p}{W_B' N_{dB}}\left(\exp\left(\frac{V_{EB}}{V_T}\right)-1\right)$$

$$= \frac{1.6\times 10^{-19}\times(10^{-4})\cdot(1.45\times 10^{10})^2}{156\times 10^{-6}\times 10^{16}}\cdot 10\cdot\left(\exp\left(\frac{0.5}{0.026}\right)-1\right)$$

$$= 2.156\times 10^{-9}\left(225\times 10^{6}\right) = 0.48\ A. \qquad (3)$$

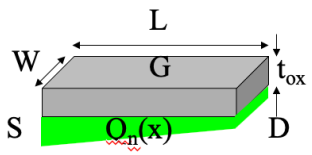
Q4:

a)



[2 + 2 +2]

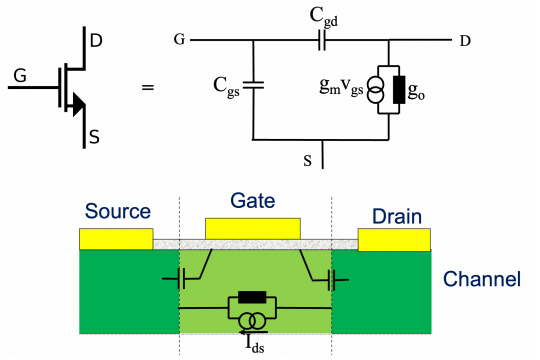
b) The small signal equivalent circuit for the MOSFET must consider capacitances associated with the change Q_n in channel charge with applied gate voltage (see diagram below). The gate capacitance may be divided into two components, C_{GS} and C_{GD}, corresponding to the change in Q_n with V_{GS} and V_{GD}.



$$C_{GS} = \frac{dQ_n}{dV_{GS}} \qquad C_{GD} = \frac{dQ_n}{dV_{GD}}$$

[2]

This gives the small signal circuit components C_{gs} and C_{gd}. Furthermore, the MOSFET small signal channel current may be approximated via a current source $i_{ds} = g_m v_{gs}$ where g_m is the transconductance. Including the channel output conductance g_0 then gives the following circuit, and locations of components within the MOSFET structure:

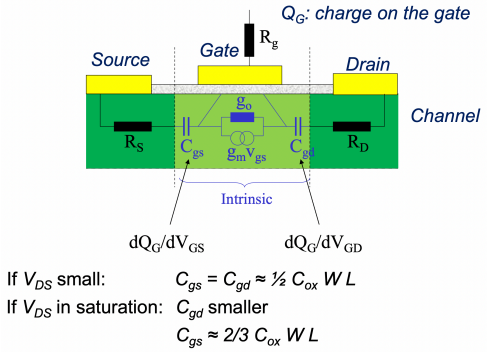


g_m is given by:

$$g_m = \frac{dI_{DS}}{dV_{GS}}$$

$$g_m^{sat} = \frac{\mu_n C_{ox} W_g}{L_g}(V_{GS} - V_{th})$$

We then have the following small signal equivalent circuit within the MOSFET structure:



If V_{DS} small: $C_{gs} = C_{gd} \approx ½\ C_{ox}\ W\ L$

If V_{DS} in saturation: C_{gd} smaller

$C_{gs} \approx 2/3\ C_{ox}\ W\ L$

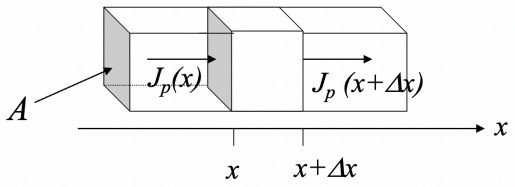
Finally, if V_{DS} is small, we may approximate $C_{gs} \sim C_{gd} \sim 0.5 C_{ox}$WL.

[4]

c)

(i)

Consider 1-D diffusion hole current flow through an infinitesimal volume AΔx, located at x. We can then write the following equation:



$$\left.\frac{\partial p}{\partial t}\right|_{x \to x+\Delta x} = \frac{1}{q}\frac{J_p(x) - J_p(x+\Delta x)}{\Delta x} - \frac{\delta p}{\tau_p}$$

Here, the LHS is the rate of hole variation, and the RHS is the sum of the variation in hole concentration in the infinitesimal volume AΔx per second, and the net recombination rate.

ANSWERS

[4]

Taking the limit for this, and using $p = p_0 + \delta p$, where p_0 is the equilibrium hole concentration, fixed by material properties and temperature:

$$\Delta x \to 0: \quad \frac{\partial p(x,t)}{\partial t} = \frac{-1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$$

$p = p_0 + \delta p$ = bulk defined + excess concentration

$$\frac{\partial \delta p}{\partial t} = \frac{-1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$$

with

$$p_0 = p_{n_0} = \frac{n_i^2}{N_D}$$

[2]

The equation for the diffusion current density J_p is:

$$J_p(x) = -eD_p \frac{dp(x)}{dx}$$

Applying this allows the required equation to be derived:

$$\frac{\partial \delta p}{\partial t} = \frac{-1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} = D_p \frac{\partial^2 p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

$p = p_0 + \delta p$ = bulk defined+ excess concentration

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

with

$$p_0 = p_{n_0} = \frac{n_i^2}{N_D}$$

[4]

(ii) Steady state conditions imply:

$$\frac{\partial \delta p}{\partial t} = 0$$

$$\therefore \frac{d^2 \delta p}{dx^2} = \frac{\delta p}{D_p \tau_p} = \frac{\delta p}{L_p^2}$$ where L_p is the diffusion length for holes.

[4]

Boundary conditions: $x = X_n \to \delta p = 0$

$x = 0 \to \delta p = \Delta p$

General solution of 2nd order differential equation: $\delta p(x) = C_1 \sinh\left(\frac{x}{L_p} + C_2\right)$

$$\delta p(x) = \frac{\Delta p}{\sinh\left(\frac{-X_n}{L_p}\right)} \sinh\left(\frac{x - X_n}{L_p}\right)$$

[4]